SANGUINE

OS9+10: Ideation and Concept Generation

Team 14A: Sanguine for Mayo Clinic
David Ricardo, Nora Alshareef, Camden Ramsay, Michel Fernandes
23 January 2026

Part 1: Ideation Workshop

0. Context

Mayo Clinic is a hospital system, including a laboratory that does tests for internal and external patients. Mayo Clinic delivers a unique "Mayo Clinic Model of Care" (MCMC) that emphasizes scientific excellence and collaboration between experts. However, for lab results to external patients, Mayo Clinic's Lab (MCL) only delivers numbers, without any of the collaboration or explanation that defines the MCMC. Our system problem statement is to improve external patient care outcomes at scale by clearly, compassionately, and collaboratively augmenting the information provided with Mayo Clinic test results using modern technology, including but not constrained to AI agents.

1. Workshop Design

1.1. What is the objective (or goal) of the workshop?

The goal of our workshop is to create new ideas for how patients get and understand their lab results. We designed this session to explore two areas: the problem space and the solution space. In the problem space, we want participants to understand what stakeholders need, find hidden assumptions, and identify the real problems that patients and healthcare workers face. In the solution space, we want to create new ideas by looking at how other industries solved similar problems and by using structured creativity methods. We also want everyone to participate and share ideas, including people with different roles and views. This includes doctors, nurses, and patients, but also patient support staff who help patients navigate the system and clinical librarians who help doctors find medical information fast. By the end, we want to have a list of ideas that come from real human needs, challenge old assumptions, learn from other industries, and use structured methods. The workshop is not about collecting old ideas. It is about creating new ones using methods like analogical thinking, assumption challenging, and structured prompts.

1.2. Innovation Methods

We structured the workshop to move through the problem space before entering the solution space. This way, we make sure ideas come from real human needs, not assumptions.

We chose **Design Thinking**[6] to anchor the workshop in the problem space. The project is about how Mayo Clinic patients get and understand their lab results. For many patients, this causes anxiety, confusion, and stress. So we need methods that focus on human feelings, not just technical systems. Patients feel scared while waiting, confused by medical words, and unsure what to do next. Design Thinking helps because it focuses on empathy. Different stakeholders like patients, doctors, nurses, and family members experience this process differently. Patients want fast access and simple explanations, but doctors want to explain results first so patients do not panic. By asking participants to take these roles and speak in first person, we find emotional problems that normal methods miss. The output is a list of prioritized pain points that shows the true problem space.

Design Thinking

Instructions for participants:
Read your Role Card aloud to your group in first person. Listen to other stakeholders and notice where needs CONFLICT and where they ALIGN. Then complete the Empathy Map for your stakeholder.

Worksheet

📋 **EMPATHY MAP WORKSHEET**
Mayo Clinic — Lab Results Experience

👤 **STAKEHOLDER:**

Who is this person?

👁 **WHAT DO THEY SEE?**

👂 **WHAT DO THEY HEAR?**

💬 **WHAT DO THEY SAY & DO?**

💭 **WHAT DO THEY THINK?**

❤️ **WHAT DO THEY FEEL?**

😣 **TOP 3 PAINS**

1.
2.
3.

⭐ **#1 Biggest Pain:**

🎯 **TOP 3 GAINS (What would make it better?)**

1.
2.
3.

⭐ **#1 Biggest Opportunity:**

💡 **IDEAS**

🚀 Tech Opportunity

🤝 Human Touch Opportunity

Completed by:

Date:

⏱️**Duration:** 10m

We included Assumption Reversal[8] to go back to the problem space and examine hidden assumptions. The lab results process has many old assumptions that people do not question. For example, doctors must see results before patients, results should be complete before sharing, and patients cannot handle raw data. These ideas exist for safety reasons, but they also cause delays and make patients feel less in control. Assumption Reversal asks participants to imagine the opposite. What if patients saw results immediately? What if they chose how much explanation they want? This helps people think beyond tradition. The output is bold ideas that challenge the status quo.

Assumption Reversal

Instructions for participants:
Identify the unwritten rules of the lab results experience — what does everyone assume must be true? Write the assumption, then FLIP it to the opposite and imagine what the experience would look like.

Template

Category	We assume that...	FLIP IT (What if the opposite?)
Timing		
Delivery		
Format		
Control		
Access		
Language		

Examples to get you started:

- Timing: We assume that doctors must see results before patients
- Delivery: We assume that bad news requires a phone call
- Format: We assume that results must be medical documents
- Control: We assume that patients cannot understand their own results

⏱️**Duration:** 10m

We used SCAMPER[7] to move fully into the solution space. SCAMPER gives structured prompts to push ideas further. What if AI helped explain results? What if results combined with appointment booking? What if the system looked like package tracking? The output is refined solution concepts ready for prototyping.

The assumptions that deserve most attention are about control, timing, and trust. The belief that physicians must control access creates delays. The idea that results should come all at once serves the organization, not patients. The belief that patients cannot understand their data underestimates modern health knowledge. We planned the workshop to first identify these assumptions in the problem space, then challenge them, and finally transform them into solutions in the solution space.

Idea Generation (SCAMPER)

Instructions for participants:

Select one bold idea from Assumption Reversal. Apply the SCAMPER prompts to transform it into new variations. Work quickly — aim for quantity over quality. Then circle your strongest 2-3 ideas.

Worksheet:

Starting Idea (from Assumption Reversal): Write your bold idea here...

Prompt	What if we...	New Idea
Substitute	replaced one part with something else?	
Combine	merged this with another service?	
Adapt	borrowed an approach from another industry?	
Modify	made it bigger, smaller, faster, slower?	
Put to other use	used this for a different purpose?	
Eliminate	removed a step or feature?	
Reverse	did the opposite or changed the order?	

Our Strongest Ideas:

1. ____________________
2. ____________________

⏱️**Duration:** 10m

1.3. Ends Analysis

Ends Analysis helps us understand goals and needs before we make solutions. It asks questions like what we want to achieve and who it is for. It does not focus on how to build things. This is important for our workshop because we need to understand the problem first. If we do not understand patients and other stakeholders, we may solve the wrong problems.

In our workshop, we explore Ends Analysis through Design Thinking and Assumption Reversal activities. From Design Thinking, we get a prioritized list of stakeholder pain points. This output shows us what patients, doctors, and families really need. For example, we discover that patients need clarity, reassurance, and faster access to results. Doctors need to prevent panic and misinterpretation. These needs become our ends. From Assumption Reversal, we get challenged assumptions that reveal hidden needs. For example, when we flip the assumption that patients cannot handle raw data, we discover a need for patient empowerment and control.

We also ask participants to think about important ilities based on these outputs. From our pain points, we identify ilities like accessibility, understandability, timeliness, and trustworthiness. For figures of merit, we use the prioritized pain points to define what success looks like. This could include time from test to patient understanding, anxiety levels, or satisfaction scores. The outputs from our problem space activities directly shape these ends.

1.4 Means Analysis

Means Analysis is relevant because it helps us think about how to create solutions after we understand the problem. It focuses on tools, methods, and technologies that can help achieve the goals. It is helpful to ideate on solutions because participants can explore different ways to solve the problem, not just one idea.

In our workshop, we explore Means Analysis through Analogical Thinking and SCAMPER activities. From Analogical Thinking, we get borrowed ideas from other industries. For example, we learn how airlines use proactive updates and how financial companies use simple dashboards. These become potential means for our solution. From SCAMPER, we get refined solution concepts. For example, we explore using AI to explain results, combining results with appointment booking, or creating interfaces like package tracking apps.

We ask participants to think about new technologies, unique combinations of existing technologies, and ways to overcome constraints. The outputs from Analogical Thinking give us inspiration for new means. The outputs from SCAMPER give us specific solution concepts that use these means. This way, our Means Analysis builds directly on what we generate in the workshop activities.

2. Personal and Team Readiness

2.1. Incorporating Readiness

We plan to meet each stakeholder’s readiness level by keeping the workshop focused and down-to-earth. This is to mitigate the risk that a participant, especially one who hasn’t enjoyed a creativity workshop before, sees the workshop as too abstract to be useful. We’re learning from how Ignacio said that the “Personal Readiness for Creativity” workshop on Wednesday the 7th had historically been “polarizing”. The workshop will specifically focus on using our limited time with each stakeholder to extract as much insight as possible, so we are preparing interview questions ahead of time.

2.2. Rehearsal

Within the team, we will rehearse in two parts. First, separate from this workshop, we will practice improvisational skills by playing theater games, like the classic “yes and” game. This exercise may seem silly to an outside stakeholder, but speaking from personal experience, does build collective improvisational skills in a small, trusted group. Then, specifically for the workshop, we will practice the agenda and questions ahead of time to ensure each team member arrives prepared. We could add even more detail, like which team members will play what role, but we can establish those details as we schedule it.

2.3. Legacy Ideas

Our main concern is that existing architectures may entrench the function-form mapping in stakeholders’ minds. Our project sponsor, an SDM alumni, is enlightened, but other stakeholders may not be. For example, it would be easy for a stakeholder to say, “standard electronic medical records systems don’t support [one particular element of form that would add information to test results], so [some function] is infeasible”. It’s important for us to respect the existing forms by being compatible with them, but we don’t want them to limit our architectures. We’ll mitigate this risk by focusing the workshop primarily on learning about desired functions, rather than forms, and referencing how our sponsor already understands this challenge.

3. Workshop Plan (one page)

3.1. Timing

We will schedule this three weeks in the future, Friday February 13. That’s the Friday before Presidents’ Day. Most people have few meetings on Fridays, so conflicts are less likely. Three weeks of advance notice will give our project sponsor enough time to organize all the other participants. Doing it in February is early enough in the semester that the OS schedule won’t expect us to lock down any forms, so we will still benefit from a function-focused workshop.

3.2. Who should be involved?

First, we analyzed all our stakeholders from Stakeholders Workshop 2026 . This table shows our classifications, initial prioritizations, importance to each other, and corresponding effort percentage from this chart:

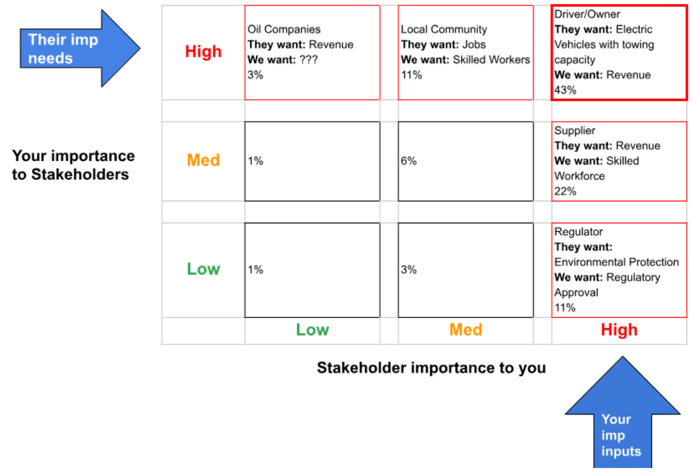


Figure 3.2.1 (above) table showing initial prioritization of stakeholders

Stakeholder	Classification	Initial prioritization	Your Importance To THEM	Their Importance to YOU	Corresponding Percentage of Effort (from Chart)
Patients	Beneficial	1	Med	High	22
Primary care physicians	Beneficial	3	Low	Med	1
Mayo Clinic Laboratories (MCL)	Beneficial	2	High	High	43
Mayo Clinic Physicians	Beneficial	2			
Mayo clinic leadership / governance	Beneficial	2	Med	Med	6
Mayo clinic legal	Problem	3	Low	High	11
Lab techs	Beneficial	3			
Software developers (at Mayo or external)	Beneficial	4	Low	Med	3
Insurance	Problem	5	Low	Med	3
Regulators (US and beyond)	Problem	3	Low	High	11

To quantify how our understanding changed over time, this chart plots the initial prioritization against the effort percentage. The top left is the most important. Stakeholders above the trendline were ones that we later realized were more important than we’d first thought, and those below the line were initially overrated. This chart also shows some low-priority stakeholders who did not make our top ten list.

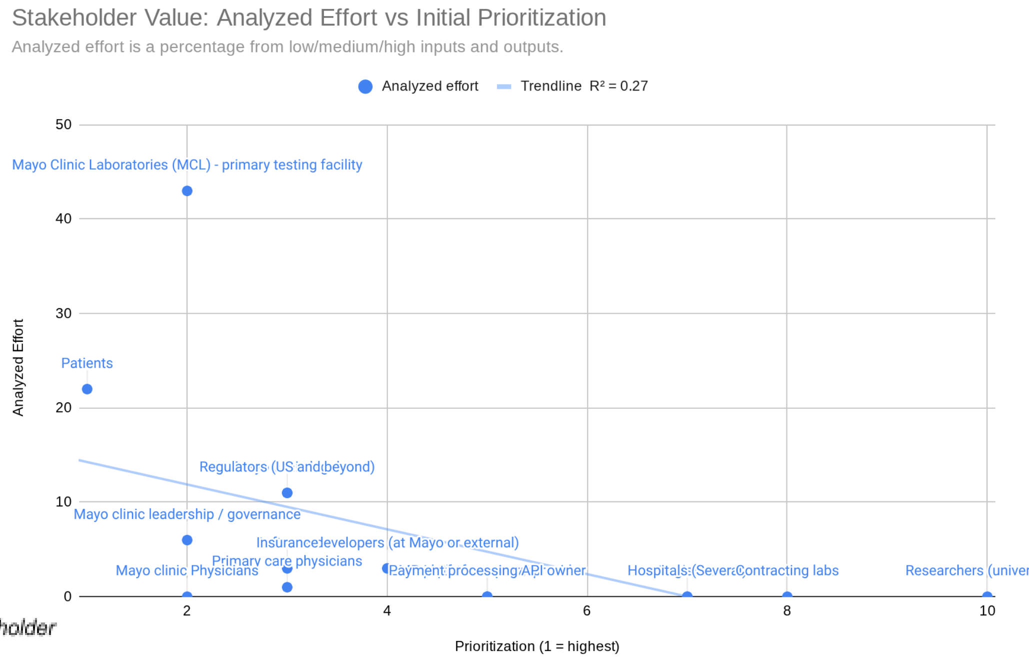


Figure 3.2.2 (right) stakeholder prioritization plot

To downselect from those stakeholders, we should invite the three most important ones. We choose three so we can do a 1:1 breakout with each of them (see 3.4 Agenda). First, we should invite one software expert in the existing lab test result delivery mechanism, to provide domain expertise on the main channel we're looking at. Second, a lab technician lead who generates results for physicians represents the source of the data pipeline. Third, we should invite a potential user: a physician within mayo clinic who has had experience ordering lab test results and could compare their experience to that of an external physician. They're the destination of the info we're building.

The main alternative we considered but discarded was a patient advocacy expert. If we could include one more stakeholder, it would be them. But we want to keep it to four stakeholders so we can do breakout interviews. We also plan to meet with legal, but they don't need to be first.

3.3. Background

Our background goals are to align participants on the existence of the problem and respect their time by asking for only limited background prereading. We will send out the one-page problem description, our system problem statement, and to make the comparison concrete, real examples of Mayo Clinic's reporting compared to competitors. Sending out the SPS beforehand will prevent stakeholders from being surprised and questioning the problem statement, which could disrupt our agenda. The concrete examples will illustrate the problem. We considered other background reading, like our project charter or OS10, but we want to minimize the stakeholder's ask.

3.4. Agenda

This table shows the agenda, including justifying each section's inclusion and duration.

Start (min)	End (min)	Activity	Description	Rationale
0	5	Introductions, agenda overview	Describe the goal of the meeting, introduce everyone, and describe the system problem statement	For a call with eight people who haven't met before, brief introductions are necessary to establish shared context.
5	10	Problem framing.	Introduce a preliminary concept of "AI librarian for physicians" as a reference point to stimulate critique and assumptions.	Starting with our non-prescriptive example will anchor the meeting to get more actionable feedback, focusing on clarification and critique.
10	20	Breakout interviews	Each participant will meet individually with one team member for an	Meeting individually lets us parallelize the discussion. Doing this before the main activity

			interview. Prepared interview questions are below.	described above mitigates groupthink by letting each stakeholder sharpen their own perspective first.
20	50	Creativity exercise	Conduct creativity activities, including empathy mapping (Design Thinking), assumption challenge (Table Flipping), and idea generation (SCAMPER).	This creativity exercise is the bulk of the workshop, so it gets as much time as everything else combined.
50	55	Wrapup	Summarizing what we learned, describing the team's action items	To keep this at only 5 minutes, our team will have to do most of the talking.
55	60	Buffer	Buffer time to make sure we don't go over.	This accounts for the inevitable "just one more thing" question and ensures we respect our stakeholders' time by ending promptly.

During the breakout interviews, we will ask these prepared questions. In ten minutes, we'll probably only get through two to four questions, so we have two questions for everyone and 1-2 more questions specialized to each stakeholder.

- Questions for everyone to ask in parallel:
  - Would you want to use this system? We need everyone to validate the concept.
  - What tools/techniques do you use to solve this problem today? This will teach us about the existing landscape and give us background to understand the other answers.
- Team member A with project sponsor will ask, "Comment on our targets from DC2 Project Charter?" This is important because a sponsor-planner misalignment could lead to retargeting the entire project later.
- Team member B with a lab tech expert will ask, "What do you wish external physicians knew when they consumed your results?"
- Team member C with a potential physician user will ask, "What information would you like to see with test results?" This helps sharpen need 2.A from Section 5.1.1 below.
- Team member D with a software expert will ask "What figures of merit (FOMs) matter most to you"? This will add a new perspective to our MOPs in section 5.2.

3.5. Instrumentation / Evaluation

We will instrument this using an AI transcription of our Zoom call. This method is simple, unobtrusive, and down-to-earth, while ensuring all the stakeholders' answers are captured. We considered alternative instrumentation techniques and discarded them. We could use a digitized post-it whiteboard tool, but that might make the workshop feel too "abstract" and risk losing the respect of stakeholders who aren't ready to buy in. Section 2.1. Incorporating Readiness, describes those concerns. We also considered a survey during or after the workshop, but quantitative data isn't meaningful at a sample size of four stakeholders.

We'll evaluate the workshop's success by the amount of meaningful learning we do during the hour. SDM team members will meet immediately after to compare results and begin synthesis. We're planning to do this workshop on a Friday (see section 3.1 Timing), so over the weekend, we'll send out a one-page summary to the stakeholders to thank them for their time, share what we learned, and invite them to follow for more.

Part 2: Operational Concept Generation

4. Context, Rationale, and Mission

4.1 Overview

The system will improve external patient care outcomes by enriching Mayo test results with additional clinical context perspectives from specialists. By introducing new ideas and viewpoints into patients' and physicians' interpretations of test results, the system aims to enhance comprehension and support decision-making. When operational, the system will enable patients and physicians to understand test results with new insights and will communicate the effectiveness and brand of the Mayo Care Model. This system will not only improve patient care outcomes but also change the perception of the Mayo brand among external patients by framing Mayo's scientifically excellent results in modern reporting.

4.2 Rationale

Existing laboratory test results often lack standardized formats and patient-specific interpretation, as each clinic may use its own structure. This lack of proper follow-up and customized interpretation can lead to misunderstandings of data unique to each patient.

On the laboratory side, results can be hard to interpret. They sometimes present unnecessary, dense data, making it challenging to understand. Data is decentralized, with each laboratory using different identifiers and, depending on location, different units.

Contextual knowledge is also hard. This knowledge comes from specialized publications or academic studies, and requires research by physicians. The absence of integrated patient-specific clinical context also limits the ability to turn laboratory test results into actionable care insights, like if a pre-existing condition explains abnormal test results.

4.2.1 Trends

Laboratory test results are delivered via web, email, or via Electronic Medical Record (EMR) platforms, like Epic MyChart[2]. This could allow patients to interpret the results on their own without a medical context, leading to misinterpretation. Recent AI advances now make it feasible to access large volumes of clinical research, which could aid interpretation.

Physicians have more work than ever, reducing their opportunity to give detailed explanations and do research for each result. This trend is reflected in Need 2.B in section 5.1.1 below. At the same time, patients are becoming more engaged in their own care, as reflected in a rise in Patient Activation Measure (PAM) levels[1][3], so patients need more interpretability (need 1.A).

Patients with chronic conditions require periodic tests over time, including different health care providers and laboratories. In these circumstances, test results are interpreted in isolation, making them less meaningful than combined results (Need 2.A).

4.2.2 Existing Systems

When results are delivered through EMRs, they only show numeric results and references to normal ranges. Historical views omit clinical events, like treatments, that could influence results. Additionally, static text tries to specific exam results measurements, but with no regard to the patient's clinical history. Patients often seek more information to understand their results from unreliable sources like internet searches.

There is still no product that offers an interpretative layer encompassing different medical perspectives and a personalized context for the patient, to support the physician's clinical judgement and provide a clear interpretation of results to patients. The proposed system aims to complement current EMR systems analyses, bridging the gap between raw results and clinically relevant interpretations that both patients and physicians clearly understand.

4.2.3 Innovation Opportunities

There are key areas within the existing laboratory processing systems that can be improved through technology. Three main areas of reducing friction or sources of error within the existing systems: ensuring data integrity and interfaces between external EMRs; analytical interpretation of test results that can provide valuable information to external patients and their physicians; and creating more compassionate and effective communication avenues to ensure information is applied to improve patient outcomes. Targeted application of agent-based specialists to facilitate the scaled MCMC across each of these modalities of the laboratory process.

Inability to connect patient profiles and EMRs across diverse, disparate systems can lead to diagnostic challenges in understanding the context of a patient's test results. Innovation through the use of intelligent systems to link patient EMRs and validate that information is correct, or to triage any detected anomalies. This agent-based approach can serve as a data specialist to ensure that patient history accessed through the EMR is complete, appropriate for that patient, and updated with test results, so that downstream systems and physicians have the required context for the diagnostic and analytical stages of patient care. Bridging this gap is a critical first step to accomplishing patient care that conforms to MCMC.

Understanding testing results, the intent that led to the lab test, and the patient's medical history is critical for properly reviewing those results within the context of the patient's medical needs. Agent-based approaches can assist in the analytical review of lab results processed by the Mayo Clinic Laboratory, with a keen understanding of medical history, prior results, intake forms, and other patient information. Such approaches must provide a clear explanation of the conclusions, with an established, shared basis in clinical research for diagnostics and for application to the specific patient. Collaboration with other specialty physicians at Mayo Clinic requires concise, sourced, and information-rich analytical reviews, leading to guidance on future patient diagnostic or treatment actions.

Communication of test results at scale to external patients without undermining the MCMC can be supported by augmenting those communication channels. This allows for continued compassion and enables physicians "time to listen, time to connect, and time to care" by leveraging agent-based approaches to facilitate dialog and post-analytical patient care, rather than replacing clinical judgment. This can lead to continued care through structured outreach and follow-up, pursuing improved external patient outcomes that better align with existing internal Mayo Clinic patient outcomes.

4.3 Missions

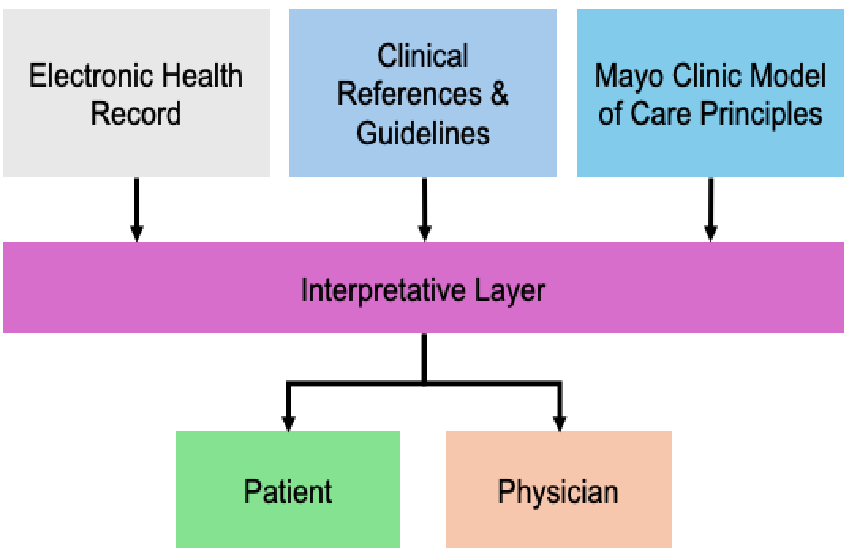


Figure 4.3 (above) showing simplified Sanguine interfaces

Primary Mission

Enable external patients and their physicians to achieve a clearer, more confident understanding of laboratory test results by providing compassionate, contextualized, and patient-specific explanatory support that augments—rather than replaces—clinical judgment.

Supporting Missions

Improve Patient Understanding and Emotional Readiness

Reduce confusion, anxiety, and misinterpretation associated with receiving laboratory test results by providing empathetic, conversational, and tailored explanations tailored to the patient's clinical context.

Augment Clinical Conversations, Not Clinical Authority

Support more productive patient-physician interactions by preparing patients with a structured understanding and informed questions, without offering diagnoses or treatment recommendations.

Support Evidence-Informed Clinical Reasoning

Provide physicians with clinical references, guidelines, and research context linked to laboratory results, supporting interpretation and informed clinical conversations without directing care decisions.

Scale the Mayo Clinic Model of Care

Extend Mayo Clinic's principles of integrated, multidisciplinary expertise to external patients through AI-enabled simulated specialist perspectives embedded in test result reporting.

5. System Needs and Performance

5.1. Needs

There are various needs of external physicians and their patients who utilize the MCL testing. The current model of providing a large “menu” of testing services with raw or commoditized results leaves patients and their physicians without assistance to provide post-analytical care. Significant clinical research, standards of care, and reference information is readily available within the Mayo Clinic. The challenge is not a lack of data available, but the curation of this information. There is a gap between test results being available for review and post-analytical care being provided to the patient; currently resolved by physicians manually bridging this gap. This disconnect or disjointed effort can lead to missed or delayed care leading to poor patient outcomes that fall short of the MCMC standards.

5.1.1 Primary Needs

(Priority is in descending order for each Primary User, when multiple needs are listed within a stakeholder section, those are in descending priority as well)

1. Patients (Primary beneficial):

1.A. Personalized health status explanation - Raw test results can be unwieldy or difficult to understand for many patients. This produces a need for concise and plain language explanations of contextual meaning derived from these results. While there often are explanatory resources available they may not be provided to the patient, or provided in a manner that is not useful to that patient. Results need to meet the patient's understanding in a manner that conforms to their needs. Translation to a local language to better serve global users, explanation in local units of measurements, and connections that are culturally adaptive, can help address these needs.

1.B. Empowered actions - Patients need to feel empowered to advocate for the care they deserve. There is a need to provide patients with simple empowering post-analytical next steps including following up with their physician, asking about what these tests mean to their care plan, and whether additional clinical diagnostic work should be scheduled can resolve this need, without providing medical advice.

1.C. Reassurance - Patients need to trust the outputs received in order to ensure long term care outcomes are not derailed due to the provision of false or incorrect information. These test results can contain information that is life changing to that patient, the care and weight this information carries must meet the patient’s emotional needs and continue to foster trust.

2. External Physicians (Primary charitable):

2.A. Specialty informing materials - External physicians often request their patients use the MCL to accommodate esoteric and specialty testing not available elsewhere[5]. These results are often linked to patient conditions that are similarly esoteric or uncommon. Mayo Clinic has reference and clinical research information available, but synthesis of where to readily find this information by the physician is needed. Connecting the physician to these contextually relevant and available resources allows for them to more confidently manage their patients care needs and next steps without the delay.

2.B. Standard workflows - Need to provide readily available standard practices of care for common treatments. This allows for information the physician already knows to be “at hand”, reducing the cognitive load from recall of memorized care steps. This also provides secondary reminders for the physician under high workload stress and can reduce research or lookup burdens and accelerate care outcomes.

3. Mayo Clinic Physicians (Primary charitable):

3.A. Bolstering existing collaborative care - While the MCMC already satisfies many of the needs facing external physicians and their patients, there is still a need to quickly provide in contextual information of internal resources to the collaborative specialist team reviewing patient results. The role of a clinical librarian is needed to curate data that these specialists rely on.

4. Mayo Clinic Leadership (Primary beneficial):

4.A. Confirm Viability - Mayo Clinic leadership has the need to ensure that the MCMC is part of their patient interactions. This can mean providing resources and insights to better serve those patients. Additionally, they have the need to ensure the financial viability of the solutions they provide so they can continue to provide best in class care to all patients they serve.

5.1.2 Non-User/Operator Needs

(Priority is in descending order for each Non-User/Operator)

5. Mayo Clinic Legal (Secondary problem):

5.A. Mitigation of liability - While the requirements (*Section 6.2)* are crafted in a manner to reduce legal liability and ensure compliance, there is a need to provide transparency for legal compliance to allow for liability assessment to Mayo Clinic’s Legal Department. The need to receive clear and documented compliance proof must be met in order to allow this architecture to satisfy primary stakeholder needs.

6. Regulators, US and beyond (Secondary problem):

6.A. Compliance Transparency - Regulators have the need to ensure that healthcare services are meeting their duty of care and providing accurate and scientifically backed clinical care.

7. Insurance providers

7.A. Reasonable Cost - While not universal globally, insurance providers have the need to ensure that care provided is prudent and reasonable in cost for reimbursement of that care.

8. Software Developers, Mayo or External (Secondary beneficial):

8.A. Defined interfaces and feedback loops - Software engineers interface with the interactive layer that coordinates test results and EMRs. Engineers need to be able to audit and test systems to ensure compliance, accuracy, and performance.

5.2 MOPs & 5.2.1. Thresholds and Objectives

Metric	Units	Description	Reasoning	Threshold value	Objective value	How it characterizes success
Patient Activation Measurement (PAM) Score [3]	Score: (0-100)	Quantifies how well prepared a patient is to manage their health.	High PAM scores are correlated with better health outcomes, so an increase in PAM would imply improved patient outcomes.	55	70	Helping patients take control of their own health is a success. This could be considered more of a MOE than a MOP, but is included here because it's linked to patient outcomes.
Net Promoter Score (NPS)	(-100 - 100)	Willingness to recommend this system.	We want users to be delighted by our system.	20	30[4]	Shows how this affects Mayo brand perception from the patient and external physicians.
Turnaround Time (TAT)	Time	Latency added by interpretation layer	Impacts whether the data is delivered quickly enough to be useful.	<12 hours	<5 minutes	Success in providing additional levels of care in a short enough duration to be useful.
Human In Loop (HIL) Escalation %	%	% of cases needing human input	Escalating to humans adds cost and latency.	90%	0%	100% independence (0% escalation) is inspiring, even if it's never achieved.
HIL TAT	Time	Turnaround time for human escalations	We know the benefits of bringing in humans, but this lets us quantify their costs.	<2 days	<24 hours	When results are delayed until human review, excessive delays would make results less useful.
Cost per test	$/test as %	Percentage additional cost per test.	More cost means less profit for Mayo Clinic. This is cost as a percentage of the test, instead of just cost in dollars, to contextualize this in existing expenses.	<10%	<5%	Impacts the NPV, a key success metric for every project.
Revenue per test	$/test as %	Percentage additional revenue per test.	Revenue is good. This metric being the percentage of the test does not commit us to charging per test; we could amortize the revenue across all the tests. This is an upper bound to avoid significantly changing the business model.	<5%	<1%	While cost can be passed to the patient or their insurance, we do want to bring in funding.

6. Operations and Requirements

6.1 Operations

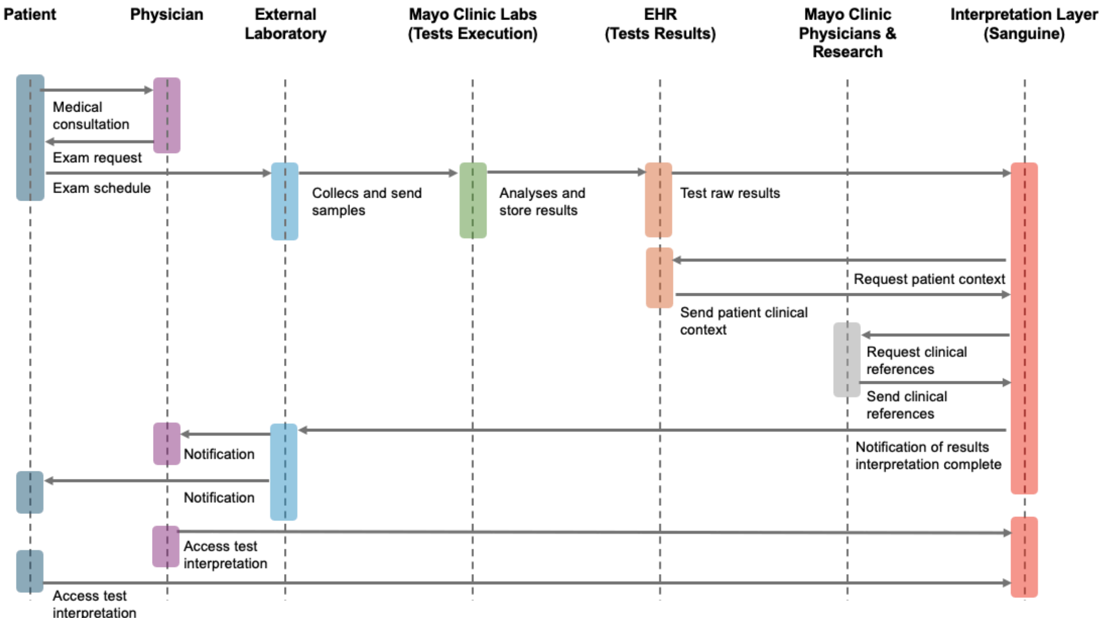


Figure 6.1 (above) Simplified operations diagram to illustrate operational flow. Simplified path shown.

This sequence diagram shows the user journey as a patient's physician requests laboratory tests. This diagram is not a set of requirements, and does not show all the existing flows.

The interpretive layer's operations are:

O1. Ingest test results

O2. Ingest additional data (possibly including patient context and clinical references)

O3. Combine results with additional data

O4. Send notifications.

6.1.1. Primary Functions

The top five functions, including traceability to the operations, are:

F1. Interface with EMR systems to consume test results (related to **O1**)

F2. Collect perspectives from Mayo Clinic physicians (**O2**)

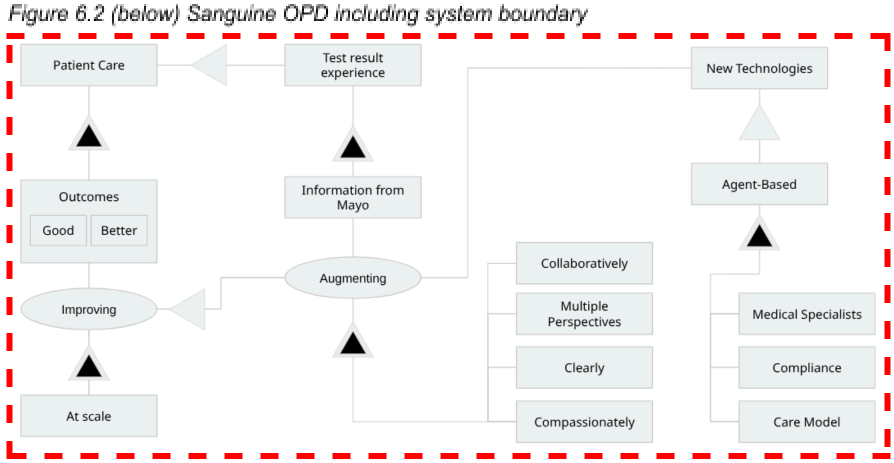
F3. Collect relevant clinical references and guidelines to support understanding for the primary care physician (**O2)**

F4. Identify the patient's persona based on the clinical context available to better explain the test results. (**O3**)

F5. Present the information with an appropriate user experience tailored to patient and physician personas (**O4**).

We originally listed many more functions, but at this early stage, we focus on the most important functions.

Figure 6.2 (below) Sanguine OPD including system boundary



6.2. Requirements

Our requirements are designed to be traceable to our needs and all the way back to our system problem statement OPD. To ensure stakeholder alignment, we have validated this problem statement and OPD with our project sponsor.

6.2.1. Functional Requirements

#	Name	Requirement text	Rationale	Traced Need
F.1	Collection of clinical perspectives	The system shall collect input from at least 1 care team member before showing results to patients	Different care team members see the process differently. Input helps coordination.	1.A, 1.C, 2.B
F.2	Patient persona identification	The system shall group patients into at least 3 health literacy levels	Systems work better when they fit each patient. Not all patients are the same.	1.A
F.3	Patient result personalization	The system shall present results in at least 2 languages, at least 2 unit systems, and at least 2 culturally adaptive formats	Results need to meet the patient's understanding in a manner that conforms to their needs, including language, units, and cultural context.	1.A
F.4	Clinical reference retrieval - cases	The system shall show at least 3 related cases to help doctors	Related cases help doctors find information fast.	2.A, 3.A
F.5	Clinical reference retrieval - sources	The system shall show at least 2 medical sources to help doctors	Medical sources help doctors learn and verify information.	2.A, 2.B
F.6	Result interpretation	The system shall show normal ranges for all	Without normal range, physicians and	1.A

	support - ranges	results	patients cannot know if result is good or bad	
F.7	Result interpretation support - trends	The system shall show trends over at least 3 past results, if available	Trends help the doctor see patterns over time.	1.A
F.8	Patient-facing result presentation	The system shall include at least 1 visual aid with each result	Visuals help patients understand results better.	1.A
F.9	Workflow integration - logins	The system shall use existing MCL sign in authentication.	physicians (and patients) reject systems that need extra work, and this also reduces costs	2.B, 5.A
F.10	Workflow integration - clicks	The system shall require no more than 2 extra clicks	Users reject systems that take too long.	2.A, 2.B
F.11	Alert management	The system shall send no more than 5 non-critical alerts per day per doctor	Too many alerts make doctors tired. They start to ignore all alerts.	2.B
F.12	Transparency of logic	Each suggestion shall be no more than 2 sentences	Doctors do not trust systems they cannot understand.	2.A, 2.B
F.13	Support for follow-up clinical conversations	The system shall provide answers to at least 10 predefined common questions per result type	Patients should not wait days for simple answers	1.B
F.14	Historic result context support	The system shall display past results or/and clinical data from at least 12 months in a timeline or graph format	Visual trends are easier to understand than numbers alone	1.A, 1.B, 1.C, 2.B
F.15	Empowered patient actions	The system shall provide at least 3 recommended next steps for each result	Patients need simple empowering post-analytical next steps to advocate for their care	1.B
F.16	Result accuracy transparency	The system shall display accuracy confidence level for 100% of generated explanations	Patients need to trust the outputs received to ensure long term care outcomes	1.C
F.17	Clinical librarian support	The system shall suggest at least 2 curated internal Mayo resources per result for specialists	Mayo specialists need quickly accessible contextual information from internal resources	3.A
F.18	Compliance audit logging	The system shall log 100% of result deliveries with timestamp and recipient	Regulators need transparency to ensure duty of care compliance	5.A, 6.A
F.19	Developer interface documentation	The system shall provide API documentation for 100% of external interfaces	Software engineers need to audit and test systems for compliance, accuracy, and performance	8.A

6.2.2. Non-functional requirements

#	Requirement name	Requirement text	Rationale	Traced Need
NF.1	Scalable interpretation throughput	The system shall be capable of generating outputs for 100 tests per day or more.	Ensure the system can operate at usage volumes without bottlenecks.	4.A. 8.A
NF.2	Multi-perspective interpretation	The system shall support at least two different medical perspectives.	Reduce interpretive bias by ensuring that each test includes different viewpoints.	1.A, 2.A
NF.3	Patient-level clarity	Outputs shall define 100% of medical jargon and acronyms used	Prevents confusion by ensuring test results don't require medical knowledge.	1.A, 1.B, 1.C
NF.4	Empathetic communication	100% of Patient-facing outputs shall acknowledge uncertainty and patient concerns when results are inconclusive or abnormal test results are present.	Mitigation of patient anxiety when the results are imprecise or with abnormal values.	1.C
NF.5	Specialists-based perspectives	Each medical perspective shall represent a different clinical specialty.	Ensure that each interpretation contributes distinctly to each test result.	2.A, 2.B, 3.A
NF.6	Mayo Clinic model alignment	The system shall align 100% of outputs with Mayo Clinic Model of Care principles	Maintains consistency of Mayo Clinic care principles.	2.A, 4.A
NF. 7	Regulatory compliance	Outputs shall only be released if 100% of validations against applicable constraints and guidelines have been passed.	Prevents non-compliant information from being released.	1.C, 4.A, 5.A, 6.A
NF. 8	Result availability notification	The system shall notify patients within 15 minutes of the results becoming available.	Reduce patient stress and anxiety by minimizing delays when results are available.	1.C
NF.9	Cost efficiency	The system shall not increase per-test cost by more than 10% of base test cost	Insurance providers need to ensure care is prudent and reasonable in cost	4.A, 7.A

Appendix: Works Cited

[0] “Testing - Insights,” Insights, Apr. 17, 2025. https://news.mayocliniclabs.com/biopharma/testing/ (accessed Jan. 23, 2026).
[1] A. Lindsay, J. H. Hibbard, D. B. Boothroyd, A. Glaseroff e S. M. Asch, “Patient activation changes as a potential signal for changes in health care costs: Cohort study of US high-cost patients,” *Journal of General Internal Medicine*, vol. 33, no. 12, pp. 2106–2112, Dec. 2018, doi: 10.1007/s11606-018-4657-6.
[2] Epic Systems Corporation, “MyChart: Patient portal,” Verona, WI, USA. [Online]. Available: https://www.mychart.com. Accessed: Jan. 2026.
[3] Hibbard, Judith H et al. “Development of the Patient Activation Measure (PAM): conceptualizing and measuring activation in patients and consumers.” Health services research vol. 39,4 Pt 1 (2004): 1005-26. doi:10.1111/j.1475-6773.2004.00269.x https://pmc.ncbi.nlm.nih.gov/articles/PMC1361049/
[4] Mayo Clinic, “Mayo Clinic NPS & Customer Reviews,” Comparably, Jan. 18, 2026. https://www.comparably.com/brands/mayo-clinic (accessed Jan. 23, 2026).
[5] “Overview - Laboratory Medicine and Pathology - Mayo Clinic,” Mayoclinic.org, 2018. https://www.mayoclinic.org/departments-centers/laboratory-medicine-pathology/overview
[6]
Interaction Design Foundation, “What is Design Thinking?,” The Interaction Design Foundation, May 25, 2016. https://www.interaction-design.org/literature/topics/design-thinking
[7] R. F. Dam and T. Y. Siang, “Scamper: How to Use the Best Ideation Methods,” The Interaction Design Foundation, Aug. 27, 2020. https://www.interaction-design.org/literature/article/learn-how-to-use-the-best-ideation-methods-scamper#scamper_method-0
[8] VanGundy, A. (1988). *Techniques of Structured Problem Solving*

Appendix: AI Use Disclosure

All ideas, concepts, and analysis in this document are the original work of the authors. AI tools (Claude/Grammerly) were used only for rephrasing and grammar checking, but all words in the submitted deliverable were produced by our team. For example, prompts included “Based on the INCOSE guide to writing requirements, which rules from section 4 does this requirement violate and why?” Gemini was used to review drafts of this assignment and act as the role of a MITsdm TA providing feedback on sections with strong arguments and areas requiring improvement. All work in response to this feedback is the author's own.